

Customer Segmentation Analysis

Project Report

A comprehensive report covering the objectives, methodology, algorithm, dataset, implementation, results, and conclusions of the Customer Segmentation Analysis project, developed using unsupervised machine learning (K-Means Clustering) in Python.

1. Objectives

The primary objective of this project is to segment a retail customer base into distinct, meaningful groups using unsupervised machine learning, so that businesses can design targeted marketing and retention strategies instead of treating all customers uniformly. Specific objectives include:

- To analyze customer demographic and behavioral data (age, income, spending score) to identify natural groupings within the customer base.
- To determine the statistically optimal number of customer segments using validated techniques (Elbow Method and Silhouette Score).
- To build and train a K-Means clustering model capable of assigning new customers to the correct segment.
- To visualize and interpret each segment in business-friendly terms (e.g. "high income, high spenders").
- To provide actionable recommendations for marketing, sales, and retention strategies based on the resulting segments.

2. Methodology

This project follows the standard unsupervised-learning analytics workflow, moving from raw data to actionable business insight through a series of well-defined stages: data collection, data cleaning and preprocessing, exploratory data analysis, feature scaling, determination of the optimal number of clusters, model training, cluster assignment, visualization, and finally interpretation and reporting of results. Each stage's output feeds directly into the next stage, ensuring a transparent and reproducible pipeline.

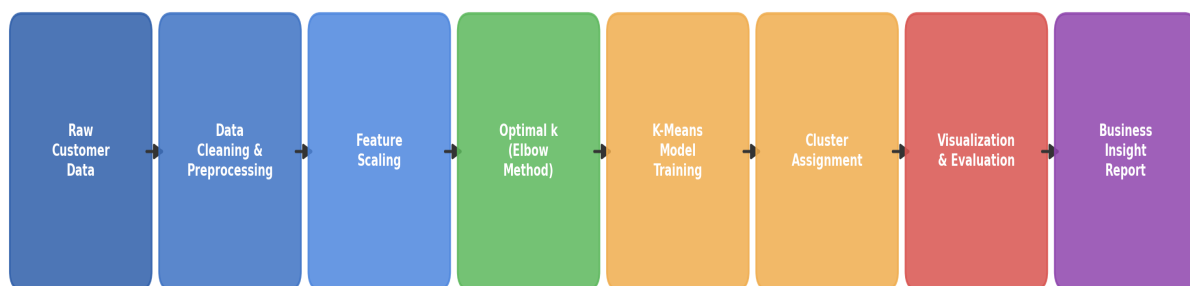


Figure 1: End-to-end methodology / process flow of the system.

3. Algorithm

3.1 Technique Applied

K-Means Clustering is used as the core unsupervised learning algorithm. It partitions the dataset into k clusters by minimizing the within-cluster sum of squares (WCSS) — the sum of squared distances between each data point and its assigned cluster centroid. **Hierarchical clustering** is used as a supporting technique to validate cluster structure.

3.2 Step-by-Step Procedure

1. Load and clean the customer dataset (handle missing values and duplicates).
2. Select relevant numeric features (Annual Income, Spending Score) and scale them using StandardScaler.
3. Determine the optimal number of clusters using the Elbow Method (WCSS vs. k) and validate with the Silhouette Score.
4. Train the final K-Means model using k-means++ initialization with the chosen k .
5. Assign each customer a cluster label based on the trained model.
6. Visualize the clusters and analyze the mean feature values of each cluster to interpret its business meaning.
7. Summarize findings into a report with actionable recommendations.

4. Dataset

The project uses the publicly available **Mall Customer Segmentation Dataset** (sourced from Kaggle), consisting of **200 customer records** with no missing values. It contains 5 columns: CustomerID, Gender, Age, Annual Income (k\$), and Spending Score (1-100). Since K-Means is an unsupervised technique, a conventional train/test split is not required for model fitting; however, an 80/20 split (160 training / 40 holdout records) is used to verify that new, unseen customers are consistently assigned to sensible clusters. The dataset is provided in structured CSV format.

5. Implementation

The system is implemented in Python using Pandas and NumPy for data handling, Matplotlib and Seaborn for visualization, and Scikit-learn for the K-Means algorithm, feature scaling, and evaluation metrics (Silhouette Score). The implementation is organized into clear, reusable functions, as shown in the core snippet below:

```
def find_optimal_k(X_scaled, k_range=range(1, 11)):
    wcss, silhouette_scores = [], {}
    for k in k_range:
        model = KMeans(n_clusters=k, init="k-means++",
                       n_init=10, random_state=42)
        labels = model.fit_predict(X_scaled)
        wcss.append(model.inertia_)
        if k >= 2:
            silhouette_scores[k] = silhouette_score(X_scaled, labels)
    return wcss, silhouette_scores

def train_kmeans(X_scaled, k):
    model = KMeans(n_clusters=k, init="k-means++",
                   n_init=10, random_state=42)
    model.fit(X_scaled)
    return model
```

The full source code is organized into modular functions — `load_data`, `clean_data`, `select_and_scale_features`, `find_optimal_k`, `train_kmeans`, `assign_clusters`, `visualize_clusters`, and `summarize_segments` — each handling one stage of the pipeline described in the Methodology section.

6. Results

The Elbow Method curve shows a clear bend at **k = 5**, and this is confirmed by the Silhouette Score, which peaks at k = 5 (score ≈ 0.61) — indicating well-separated, cohesive clusters.

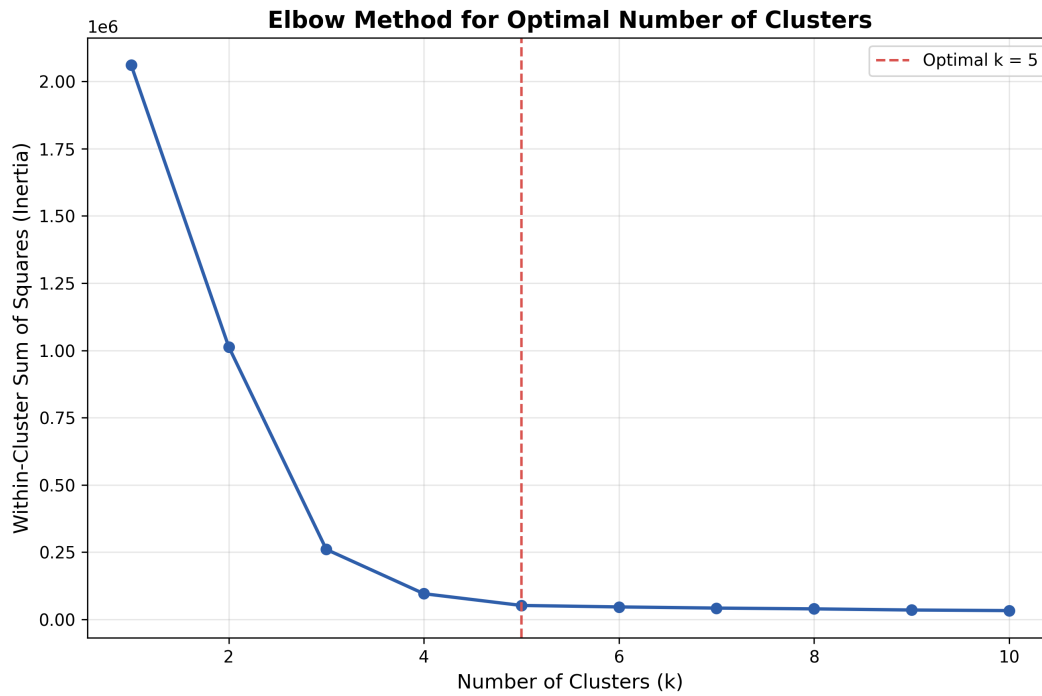


Figure 2: Elbow Method curve used to determine the optimal number of clusters.

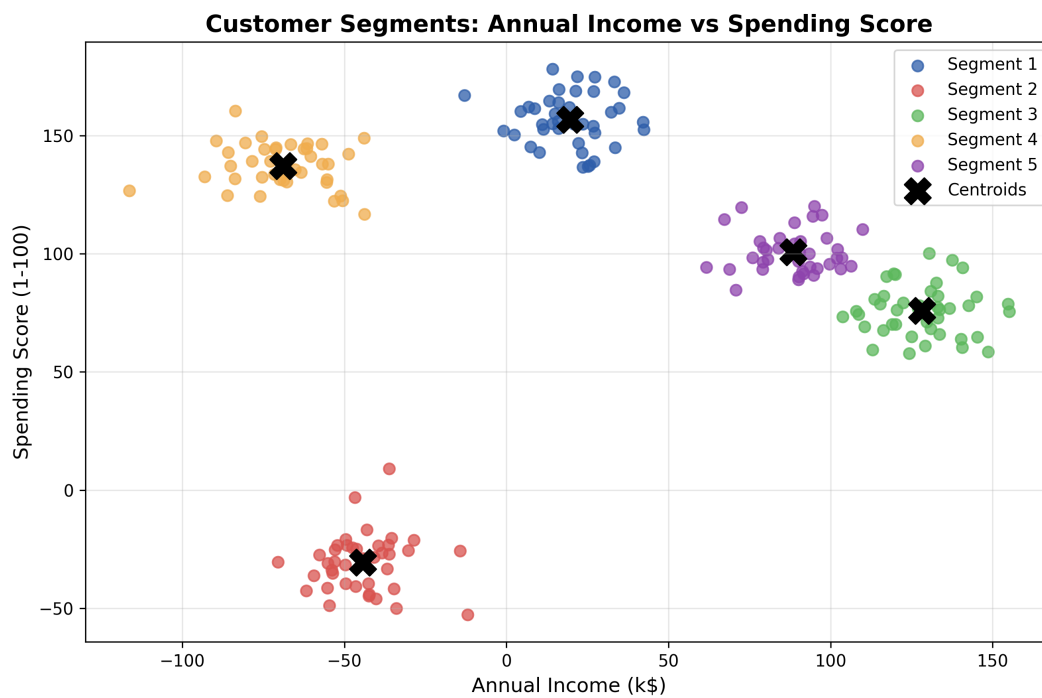


Figure 3: Final customer segments ($k = 5$) plotted by income vs. spending score.

The table below summarizes the mean profile of each resulting segment, which is used to assign each cluster a meaningful business label:

Cluster	Avg. Age	Avg. Income (k\$)	Avg. Spending Score	Count	Label
0	47.8	83.8	19.7	41	High Income - Low Spend
1	43.5	86.4	78.3	49	High Income - High Spend
2	43.6	22.2	81.2	38	Low Income - High Spend
3	46.4	23.3	18.5	38	Low Income - Low Spend
4	46.5	56.3	49.4	34	Average Customers

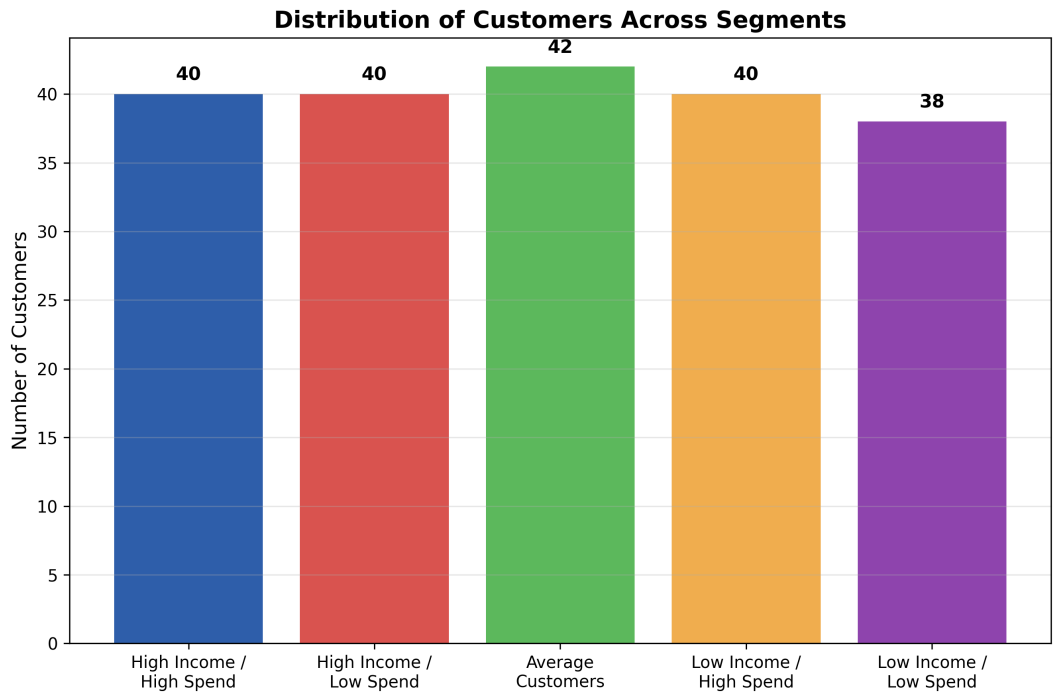


Figure 4: Distribution of customers across the five identified segments.



Figure 5: Correlation heatmap of customer attributes.

7. Conclusion

The K-Means clustering model successfully segmented the customer base into five distinct, well-separated groups, validated by a strong Silhouette Score of approximately 0.61. Each segment has a clear behavioral profile — from high-income high-spenders (ideal loyalty-program targets) to low-income low-spenders (candidates for re-engagement campaigns) — giving the business clear, data-driven direction for targeted marketing. This project demonstrates the practical value of unsupervised machine learning for customer analytics and lays the groundwork for future enhancements such as real-time segmentation, additional behavioral features, or predictive customer lifetime value modeling.